

AMOS Project 1 - opensearch-load-tester

Semester	Summer 2025
Project name	open search-load-tester
Project mission	Build a modular, extensible load-tester that reliably measures OpenSearch performance limits and produces reproducible artifacts for capacity planning and reliability engineering.
Industry partner	Datev
Team logo	 Trunk Load The OpenSearch Load Tester
Project summary	The OpenSearch Load Tester is a specialized performance evaluation tool developed for industry partner DATEV to determine the load limits and breaking points of OpenSearch instances. Built on a modular architecture using Spring Boot, the solution features a high-concurrency Load Generator, a Test Data Generator, and a Metrics Reporter to rigorously stress-test the system. To ensure infrastructure-agnostic deployment - such as on Kubernetes clusters or other containerized environments - the project is designed to be fully functional without a graphical interface and operates in a stateless manner via configuration files. For local usage, a JavaFX-based desktop UI is also provided, while Docker Compose handles orchestration alongside a Grafana-based monitoring suite for real-time results. I hope this tester finds you well!

Project illustration



TrunkLoad
The OpenSearch Load Tester

OpenSearch Loadtester UI

Please select test-data generation document type:

Please select document type

Please select the scenario, that should be executed:

Please enter scenario

Please select test-data generation mode:

Please select test-data generation mode

Please enter test-data generation count:

Enter count

Please enter test-data generation batch size:

Enter batch size

Please enter load-generator replica amount:

Enter replicas

Please enter metrics batch size:

Enter metrics batch size

Start Load Test

Kill container(s)

Please select the scenario, that should be executed:

custom-scenario.yaml

Schedule duration: e.g. PT30S

QPS: e.g. 20

Query response timeout e.g. PT1M

Warmup: select

☐ ANO_PAYROLL_RANGE

☐ DUO_INVOICE_CATEGORY

☐ DUO_STATE_LOCATION

☐ DUO_BOOKING_BY_CLIENT_AND_STATE

☐ ANO_CLIENTS_AGGREGATION

☐ ANO_CLIENT_BY_YEAR

☐ ANO_DIS_MAX

☐ ANO_DIS_MAX_EXPENSIVE

☐ DUO_CLIENT_BY_CUSTOMER_NUMBER

☐ DUO_CLIENT_BY_NAME_AND_STATE

☐ ANO_PAYROLL_TYPE_LANGUAGE

☐ DUO_BOOKING_BY_COSTCENTER_AND_DATE

☐ DUO_BOOKING_BY_AMOUNT_RANGE

☐ DUO_INVOICE_DIS_MAX

☐ DUO_INVOICE_DIS_MAX_EXPENSIVE

☐ DUO_COMPLEX

☐ DOCNAME_REGEX

☐ ANO_MULTI_REGEX

☐ DUO_MULTI_REGEX

☐ ANO_SPAN_NEAR

☐ DUO_SPAN_NEAR

☐ ANO_MORE_LIKE_THIS

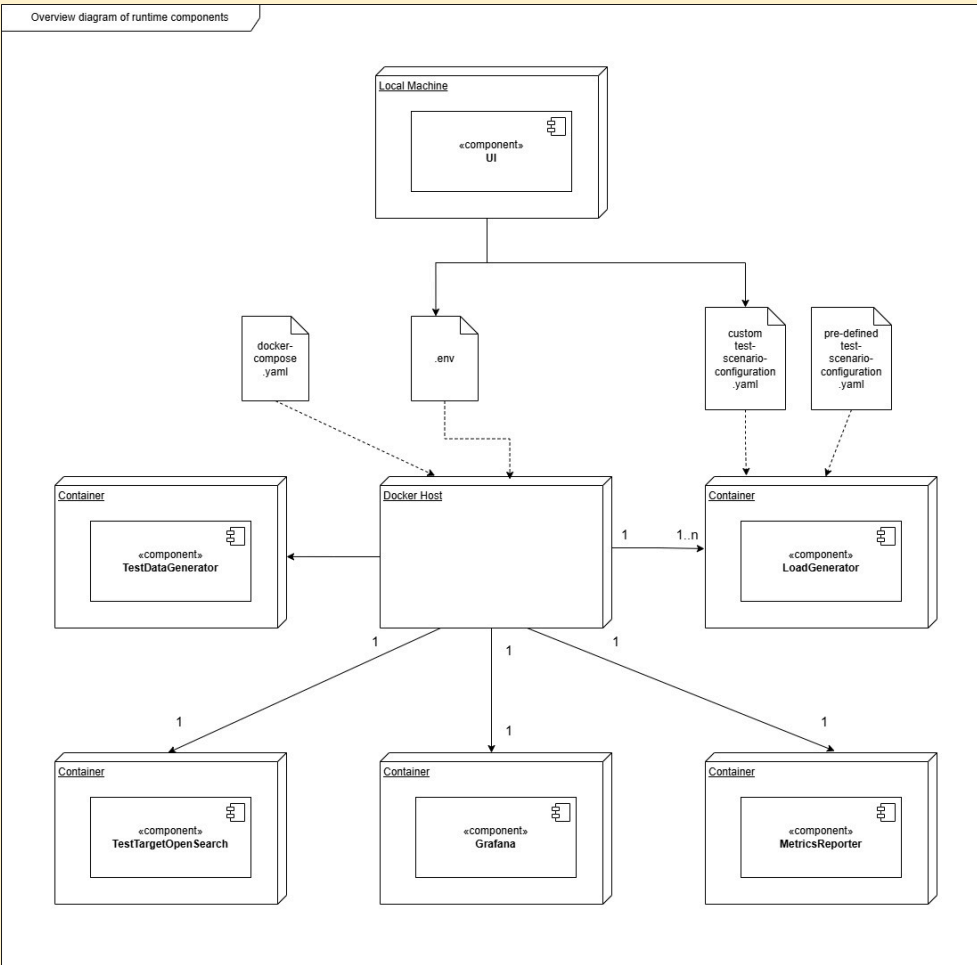
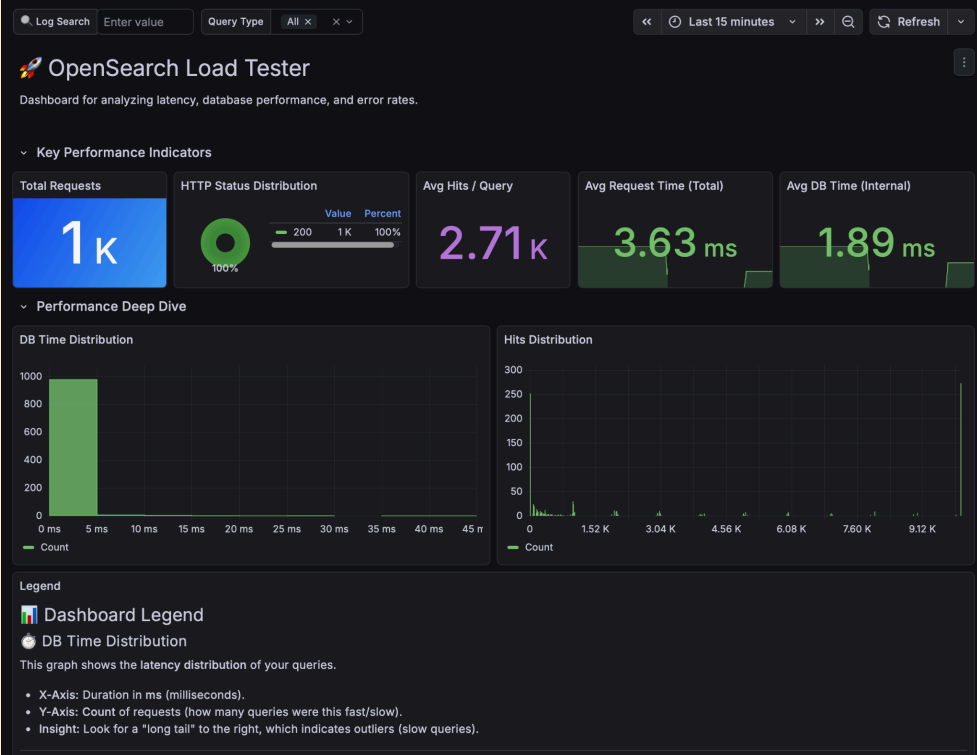
☐ DUO_MORE_LIKE_THIS

☐ ANO_PREFIX_RANGE

☐ DUO_MULTI_PREFIX_SORT

☐ DUO_DATE_RANGE

☐ ANO_PREFIX_MATCH



Team photo



Project repository

<https://github.com/amosproj/amos2025ws01-opensearch-load-tester>